30	**C**	The energy of the photon emitted corresponds to the transitions between the energy levels in the hydrogen atom. $\Delta E = \dfrac{hc}{\lambda} = \dfrac{(6.63 \times 10^{-34})(3.0 \times 10^{8})}{656.28 \times 10^{-9}} = 3.03 \times 10^{-19}\,\text{J} = 1.9\ \text{eV (2sf)}$ Option A : ΔE = (-3.4)-(-13.6) = 10.2 eV Option B : ΔE = (-1.5)-(-13.6) = 12.1 eV Option C : ΔE = (-1.5)-(-3.4) = 1.9 eV (Answer) Option D : ΔE = (-0.5)-(-3.4) = 2.9 eV <u>Alternative method (without calculation)</u> It should be noted that visible light range occurs for the Balmer series where the transition involves n = 2. So the answer is narrowed to C or D. Red light has the longest wavelength and so the energy transition must be the smallest. Likely n = 3 to n = 2.